FAULT CODE 1548 (QSK50 and QSK60) - Injector Solenoid Driver Cylinder 7 Circuit - Current Below Normal or Open Circuit

Warnings and Cautions

⚠ WARNING ⚠

The injector solenoids receive high voltage when the engine is operating. To reduce the possibility of personal injury from electrical shock, do not wear jewelry or damp clothing, and do not touch the injector solenoids or the solenoid wires when the engine is operating.

⚠ CAUTION ⚠

To reduce the possibility of damaging a new engine control module (ECM), all other active fault codes must be investigated prior to replacing the ECM.

⚠ CAUTION ⚠

To reduce the possibility of pin and harness damage, use the following test leads when taking a measurement: Part Number 3824811 - male Deutsch™ test lead, Part Number 3824812 - female Deutsch™ test lead, Part Number 3822758 - male Deutsch™/AMP™/Metri-Pack™ test lead and Part Number 3822917 - female Deutsch™/AMP™/Metri-Pack™ test lead.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check for active fault codes.	
	STEP 1A. Read the fault codes with INSITE™ electronic service tool.	Fault Code 1548, 325, or 1555 active?
	STEP 1B. Read the fault codes with INSITE™ electronic service tool.	**Only** Fault Code 1548 is active?

STEPS		SPECIFICATIONS
	STEP 1C. Read the fault codes with INSITE™ electronic service tool.	Multiple injector fault codes active?
STEP 2.	Check the injector solenoid driver cylinder 7 circuit for an open circuit.	
	STEP 2A. Inspect the engine harness connections.	Connectors properly connected?
	STEP 2A-1. Inspect the engine harness and ECM connector pins.	Dirty or damaged pins?
	STEP 2B. Check for an open circuit.	Resistance between 0.5 and 5 ohms?
	STEP 2C. Inspect the injector connector pins.	Dirty or damaged pins?
	STEP 2D. Check for an open circuit.	Resistance between 0.5 and 5 ohms?
	STEP 2E. Read the fault codes.	**Only** Fault Code 1548 is active?
STEP 3.	Check the engine harness.	
	STEP 3A. Inspect the engine harness and injector connector pins.	Dirty or damaged pins?
	STEP 3B. Check the injector solenoids for short circuits to ground.	Greater than 100k ohms?
	STEP 3C. Inspect the engine harness.	Dirty or damaged pins, or damaged wire insulation?
	STEP 3C-1. Check the engine harness for a short circuit to ground.	Greater than 100k ohms?
	STEP 3C-2. Check the engine harness for a pin-to-pin short circuit.	Greater than 100k ohms?
STEP 4.	Disable and clear the fault codes.	
	STEP 4A. Disable the fault code.	Same multiple injector fault codes active?
	STEP 4B. Clear the inactive fault codes.	All fault codes cleared?

STEP 1. Check for active fault codes.

STEP 1A. Read the fault codes with INSITE™ electronic service tool.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Start the engine and let it idle for 1 minute. • Use INSITE™ electronic service tool to clear the fault codes.	Fault Code 1548, 325, or 1555 active? **YES**	**1B**
	Fault Code 1548, 325, or 1555 active? **NO**	**Use the following procedure for inactive or intermittent fault code. Refer to Procedure 019-362 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-362.html)**

STEP 1B. Read the fault codes with INSITE™ electronic service tool.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Start the engine and let it idle for 1 minute. • Use INSITE™ electronic service tool to read the fault codes.	**Only** Fault Code 1548 active? **YES**	**2A**
	Only Fault Code 1548 active? **NO**	**1C**

STEP 1C. Read the fault codes with INSITE™ electronic service tool.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Start the engine and let it idle for 1 minute. • Use INSITE™ electronic service tool to read the fault codes.	Multiple injector fault codes active? **YES**	**3A**
	Multiple injector fault codes active? **NO**	**2A**

STEP 2. Check the injector solenoid driver cylinder 7 circuit for an open circuit.

STEP 2A. Inspect the engine harness connections.

Conditions :
- Turn keyswitch OFF.

Action	Specification/Repair	Next Step
Make sure the following engine harness connections are properly made: • Engine harness connected to ECM • Engine harness connected to the injector solenoid driver cylinder 7.	Connectors properly connected? **YES**	**2A-1**
	Connectors properly connected? **NO** **Repair :** Install the engine harness connectors properly.	**4A**

STEP 2A-1. Inspect the engine harness and ECM connector pins.

Conditions :
- Turn keyswitch OFF.
- Disconnect the engine harness connector from the ECM 60 pin connector.

Action	Specification/Repair	Next Step

Inspect the engine harness and ECM connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? **YES** **Repair :** Clean the connector and pins. Replace the damaged section of the harness. Refer to the circuit diagram or wiring diagram for all harness interconnections. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html)	**4A**
	Dirty or damaged pins? **NO**	**2B**

STEP 2B. Check for an open circuit.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness connector from the ECM 60 pin connector.

Action	Specification/Repair	Next Step

Check for continuity in the injector solenoid driver cylinder 7 circuit. • Measure the resistance between the injector solenoid driver cylinder 7 SIGNAL pin and the injector solenoid driver cylinder 7 RETURN pin at the ECM 60 pin connector of the engine harness. Refer to the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Resistance between 0.5 and 5 ohms? **YES**	**2E**
	Resistance between 0.5 and 5 ohms? **NO**	**2C**

STEP 2C. Inspect the injector connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness connector from the ECM 60 pin connector.
- Disconnect the engine harness connector from the injector solenoid driver cylinder 7 connector.

Action	Specification/Repair	Next Step

Inspect the engine harness and injector solenoid driver cylinder 7 connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire or insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? **YES** **Repair :** Clean the connector and pins. Replace the damaged section of the harness or damaged injector. Refer to the circuit diagram or wiring diagram for all harness interconnections. Repair the engine harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html) Replace the damaged injector. • Use the following procedure from the K38, K50, QSK38 and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html) • Use the following procedure from the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)	**4A**

	Dirty or damaged pins? **NO**	**2D**

STEP 2D. Check for an open circuit.

Conditions :

- Turn the keyswitch OFF.
- Disconnect the engine harness connector from the ECM 60 pin connector.
- Disconnect the engine harness connector from the injector solenoid driver cylinder 7 connector.

Action	Specification/Repair	Next Step

Check for continuity in the injector solenoid driver cylinder 7. • Measure the resistance between the injector solenoid driver cylinder 7 SIGNAL pin and the injector solenoid driver cylinder 7 RETURN pin at the injector solenoid driver cylinder 7 connector. Refer to the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Resistance between 0.5 and 5 ohms? **YES** **Repair :** Troubleshoot all harnesses connected in series to determine which contains the open circuit. Refer to the circuit diagram or wiring diagram for all harness interconnections. Replace the damaged section of the harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html)	**4A**

	Resistance between 0.5 and 5 ohms? **NO** **Repair :** Replace the damaged injector. • Use the following procedure from the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html) • Use the following procedure from the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)	**4A**

STEP 2E. Read the fault codes.

Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect the INSITE™ electronic service tool.

Action	Specification/Repair	Next Step

Operate the engine and observe the fault codes. • Start the engine and let it idle for 1 minute. • Use INSITE™ electronic service tool to clear the fault codes.	**Only** Fault Code 1548 is active? **YES** **Repair :** Replace the ECM. Refer to Procedure 019-031 in Section 19. (/qs3/pubsys2/xml/en/procedures/105/105-019-031.html)	**4A**
	Only Fault Code 1548 is active? **NO**	**4A**

STEP 3. Check the engine harness.

STEP 3A. Inspect the engine harness and injector connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the injector solenoid driver cylinder 7, 6, and 14 connectors from the engine harness connectors.

Action	Specification/Repair	Next Step

<table>
<tr>
<td rowspan="2">Inspect the engine harness and injector solenoid driver cylinder 7, 6, and 14 connector pins for the following:
Loose connector
Corroded pins
Bent or broken pins
Pushed back or expanded pins
Moisture in or on the connector
Missing or damaged connector seals
Dirt or debris in or on the connector pins
Connector shell broken
Wire or insulation damage
Damaged connector locking tab.
Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)</td>
<td>Dirty or damaged pins?
YES

Repair :
Clean the connector and pins.
Replace the damaged section of the harness or damaged injector(s).
Refer to the circuit diagram or wiring diagram for all harness interconnections.
Repair the engine harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html)
Replace the damaged injector.
Use the following procedure from the K38, K50, QSK38 and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html)
Use the following procedure from the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)</td>
<td>4A</td>
</tr>
<tr>
<td>Dirty or damaged pins?
NO</td>
<td>3B</td>
</tr>
</table>

STEP 3B. Check the injector solenoids for a short circuit to ground.

Conditions :

- Turn the keyswitch OFF.
- Disconnect the injector solenoid driver cylinder 7, 6, and 14 connectors from the engine harness connectors.

Action	Specification/Repair	Next Step
Check for a short circuit to ground. • Measure the resistance between the injector solenoid driver cylinder 7 SIGNAL pin and engine block ground. • Measure the resistance between the injector solenoid driver cylinder 6 SIGNAL pin and engine block ground. • Measure the resistance between the injector solenoid driver cylinder 14 SIGNAL pin and engine block ground. Refer to the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Greater than 100k ohms? **YES**	**3C**
	Greater than 100k ohms? **NO** **Repair :** Replace the damaged injector(s). Use the following procedure from the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html) Use the following procedure from the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)	**4A**

STEP 3C. Inspect the engine harness.

Conditions :

- Turn keyswitch OFF.
- Disconnect the injector solenoid driver cylinder 7, 6, and 14 connectors from the engine harness connectors.
- Disconnect the engine harness connector from the ECM 60 pin connector.

Action	Specification/Repair	Next Step

<table>
<tr>
<td rowspan="2">Inspect the engine harness and ECM connectors for the following:
Loose connector
Corroded pins
Bent or broken pins
Pushed back or expanded pins
Moisture in or on the connector
Missing or damaged connector seals
Dirt or debris in or on the connector pins
Connector shell broken
Wire or insulation damage
Damaged connector locking tab.
Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)</td>
<td>Dirty or damaged pins, or damaged wire insulation?
YES
Repair :
Replace the damaged section of the harness or damaged injector(s)
Refer to the circuit diagram or wiring diagram for all harness interconnections.
Repair the engine harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html)
Replace the damaged injector.
Use the following procedure from the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html)
Use the following procedure from the QSK45 and QSK60 Service Manual, Bulletin 4021530. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)</td>
<td>4A</td>
</tr>
<tr>
<td>Dirty or damaged pins, or damaged wire insulation?
NO</td>
<td>3C-1</td>
</tr>
</table>

STEP 3C-1. Check the engine harness for a short circuit to ground.

Conditions :		
- Turn the keyswitch OFF. - Disconnect the engine harness connector from the ECM 60 pin connector. - Disconnect the injector solenoid driver cylinder 7, 6, and 14 connectors from the engine harness connectors.		
Action	**Specification/Repair**	**Next Step**

Check for a short circuit to ground. - Measure the resistance from the injector solenoid driver cylinder 7 SIGNAL pin at the ECM 60 pin connector of the engine harness to engine block ground. Repeat the check at the injector solenoid driver cylinder 6 SIGNAL and injector solenoid driver cylinder 14 SIGNAL pins. - Measure the resistance from the injector solenoid driver cylinder 7 RETURN pin at the ECM 60 pin connector of the engine harness to engine block ground. Repeat the check for the injector solenoid driver cylinder 6 RETURN and injector solenoid driver cylinder 14 RETURN pins. Refer to the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Greater than 100k ohms? **YES**	**3C-2**
	Greater than 100k ohms? **NO** **Repair :** Troubleshoot all harnesses connected in series to determine which contains the short circuit. Refer to the circuit diagram or wiring diagram for all harness interconnections. Replace the damaged section of the harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html)	**4A**

STEP 3C-2. Check the engine harness for a pin-to-pin short circuit.

Conditions :

- Turn the keyswitch OFF.
- Disconnect the engine harness connector from the ECM 60 pin connector.
- Disconnect the injector solenoid driver cylinder 7, 6, and 14 connectors from the engine harness connectors.

Action	Specification/Repair	Next Step

Check for a pin-to-pin short circuit. • Measure the resistance from the injector solenoid driver cylinder 7 SIGNAL pin at the ECM 60 pin connector of the engine harness to all other pins in the connector. Repeat the check at the injector solenoid driver cylinder 6 SIGNAL and injector solenoid driver cylinder 14 SIGNAL pins. • Measure the resistance from the injector solenoid driver cylinder 7 RETURN pin at the ECM 60 pin connector of the engine harness to all other pins in the connector. Repeat the check for the injector solenoid driver cylinder 6 RETURN and injector solenoid driver cylinder 14 RETURN pins. Refer to the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Greater than 100k ohms? **YES**	**4A**
	Greater than 100k ohms? **NO** **Repair :** Troubleshoot all harnesses connected in series to determine which contains the pin-to-pin short. Refer to the circuit diagram or wiring diagram for all harness interconnections. Replace the damaged section of the harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html)	**4A**

STEP 4. Disable and clear the fault codes.

STEP 4A. Disable the fault code.

Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Disable the fault code. • Start the engine and let it idle for 1 minute. • Use INSITE™ electronic service tool to verify that the fault codes are inactive.	Same multiple injector fault codes active? **YES** **Repair :** Verify that all steps have been completed. If all steps have been completed, then follow the technical escalation process.	**Escalate or call for assistance.**
	Same multiple injector fault codes active? **NO**	**4B**

STEP 4B. Clear the inactive fault codes.

Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Clear the inactive fault codes. • Use INSITE™ electronic service tool to clear the inactive fault codes.	All fault codes cleared? **YES**	**Repair complete.**
	All fault codes cleared? **NO**	**1A**

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